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PRN No: 202501040030
4.1.1. Pandas - series creation and manipulation

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Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

```
# import pandas as pd

# # Take inputs from the user to create a list of numbers
# numbers = list(map(int, input().split()))

# grouped =

# # Display the mean of even and odd numbers with labels
# grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
# print("Mean of even and odd numbers:")
# print(grouped)
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
s = pd.Series(numbers)

# Grouping by even and odd numbers and calculating the mean
grouped = s.groupby(s % 2 == 0).mean()

# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]

print("Mean of even and odd numbers:")
```

print(grouped)